

NGUYEN KHAC BAO

AI Engineer Intern – LLM/RAG & ERP Workflow Integration

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CAREER OBJECTIVE

AI Engineer Intern candidate focused on building practical AI tools for business operations, ERP-style workflows, document processing, reporting, and chatbot automation. Hands-on with Python, FastAPI, REST APIs, Streamlit, Gemini API, RAG chatbot, SQL/vector databases, Docker, and cloud deployment.

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT)

2022 – 2027

Bachelor of Information Technology

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision.

TECHNICAL SKILLS

Programming: Python, SQL basics, JavaScript basics

LLM / RAG: Gemini API, OpenAI-compatible APIs, RAG pipelines, embeddings, vector search, prompt engineering

Backend / API: FastAPI, REST API, Streamlit, JSON/API integration

Data / SQL: Pandas, CSV/JSONL processing, MySQL/Cloud SQL, MariaDB, Qdrant Cloud

DevOps / Cloud: Docker, Linux, Git/GitHub, Google Cloud Run, AWS S3/SageMaker, Vast.ai, Hugging Face

AI / CV / OCR: OpenCV, OCR, YOLOv8n, object detection, dataset QA

LANGUAGES

Vietnamese: Native | English: TOEIC 610/990; technical reading proficiency for documentation, API references, model cards, and engineering articles.

PROJECTS

Vietnamese Legal RAG Chatbot

2026

Personal Project | Python, FastAPI, Streamlit, Gemini API, Qdrant Cloud, Cloud SQL, Docker, Google Cloud Run | Source: [GitHub](#)

- Built and deployed a **RAG chatbot** with **Streamlit** frontend and **FastAPI** backend as separate **Cloud Run** services.
- Indexed 32,980 Vietnamese QA/context records into **Qdrant Cloud** using **Gemini embeddings** for retrieval-grounded answers.
- Implemented **chat APIs**, **Cloud SQL** chat history, and Secret Manager-based credential handling.
- Designed the architecture to support internal documents, SQL-backed records, ERP-style workflows, async indexing, and optional OpenAI-compatible LLM serving.

AutoFall Labeler Tool

2026

Personal Project | Python, Streamlit, OpenCV, YOLO-Pose, Pandas | Source: [GitHub](#)

- Built a **human-in-the-loop workflow tool** with Streamlit review UI, metadata/annotation management, YOLO-Pose keypoint suggestions, and reviewed-only export.
- Added evaluation reports for **auto-label accuracy**, correction rate, pose missing rate, and keypoint edit rate.
- Designed CSV-based source-of-truth records for annotations, metadata, keypoints, and review events to support reproducible workflow documentation.

Mini Microservices Social Web

2026

Personal Project | React, Node.js, Express Gateway, RabbitMQ, PostgreSQL/Cloud SQL, Docker Compose | Source: [GitHub](#)

- Built a mini social web system with **API Gateway**, user/post/comment/feed/notification services, JWT-protected routes, and **RabbitMQ** event messaging.
- Containerized frontend, gateway, backend services, RabbitMQ, and database configuration using **Docker Compose**.

Traffic Violation Detection & License Plate Recognition

2026

Personal Project | Python, OpenCV, YOLO, OCR, JSON | Source: [GitHub](#)

- Built an AI video analytics pipeline for vehicle detection/tracking, traffic-light state estimation, stop-line violation logic, **OCR**, and JSON evidence generation.
- Fine-tuned a YOLO plate detector on 8,598 CCPD images, achieving 0.989 precision, 0.998 recall, and 0.994 mAP50.

RELEVANT STRENGTHS

- Built practical AI automation tools using Python, Streamlit/FastAPI, API integration, and Docker-based deployment.
- Comfortable processing CSV/JSON data, designing metadata schemas, generating QA reports, and documenting workflow outcomes.
- Demonstrated problem-solving through real-world CV/OCR pipelines involving tracking, noisy data, edge cases, and human-in-the-loop validation.

ACTIVITIES

AI Research Group – University Club

2025 – Present

Member

- Discussed AI applications, workflow automation ideas, Machine Learning, Deep Learning, Computer Vision, and RAG/LLM systems.